

## SEMESTER S5

### ADVANCED COMPUTER ARCHITECTURE

|                                            |                 |                    |                |
|--------------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PECST528</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | PBCST404        | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To introduce the advanced processor architectures including parallelism concepts in Programming of multiprocessor and multicomputers.
2. To provide detailed understanding about data flow in computer architectures.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <p>Introduction – The impact of hardware and software technology trends Self review – Instruction set Architecture, Memory addressing, addressing modes Class of Computers, Concept of Computer Hardware and Organization (P15, 5th Edition) Measuring, Reporting and Summarizing Performance, Benchmarks – Desktop and Server Amdahl's Law, Processor Performance Equation</p> <hr/> <p><i>Beyond the books</i> – Visit <a href="http://www.spec.org">www.spec.org</a>. Explore the High Performance Computing benchmarks and compare the results submitted by different vendors for the same benchmark. Are you able to appreciate the need for benchmarks to compare performance? What are retired benchmarks? Can you write a paper and publish results based on a retired benchmark?</p> |                      |
| <b>2</b>          | <p>Review the basic Concepts of Parallel Processing and Pipelining Instruction Level Parallelism, data dependencies and hazards Different types of dependences, Compiler Techniques for ILP, Branch Prediction – Correlating</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                      |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|          | branch predictor Dynamic Scheduling – Idea, Introduction to Tomasulo’s scheme. Register Renaming Hardware Speculation, Reorder Buffers Multiple issue and static scheduling, VLIW                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
| <b>3</b> | Data Level Parallelism. Vector Processors – How do they work, Memory Banks, Stride, Scatter Gather. SIMD-comparison with vector GPU, Comparison of loops in C vs CUDA NVIDIA GPU Memory structure Vector Processor vs GPU, Multimedia SIMD computers vs GPU Multiprocessor Architecture, Centralized shared memory architecture Cache coherence and snooping protocol (Implementation details – not required). Performance of Symmetric Shared-Memory Processors. Distributed Shared Memory and Directory based protocol – basics. Synchronization – Basic Hardware Primitives. Memory Consistency Models – Sequential and relaxed                                                                                                        |  |
| <b>4</b> | Warehouse Scale Computers – Goals and requirements. Programming frameworks for Batch processing – Map reduce and Hadoop Computer Architecture of Warehouse-scale computers Moore’s Law, Dennard Scaling, Dark Silicon and the transition towards Heterogeneous Architectures Asymmetric multi-core architecture – Static and Dynamic (Overall idea, example processors) Functional Heterogeneous Multicore architecture – GPUs, Accelerators, Reconfigurable Computing Beyond the textbook – Identify the processor used in your PC and mobile phone. Study about its architecture, is it homogeneous or heterogeneous, does it use GPUs, what information can you gather about it from the manufacturer’s website – Discuss in the class |  |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

### End Semester Examination Marks (ESE)

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                              | Part B                                                                                                                                                                                                                                                                            | Total     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"><li>• 2 Questions from each module.</li><li>• Total of 8 Questions, each carrying 3 marks</li></ul> <p><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"><li>• Each question carries 9 marks.</li><li>• Two questions will be given from each module, out of which 1 question should be answered.</li><li>• Each question can have a maximum of 3 subdivisions.</li></ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                             | Bloom's Knowledge Level (KL) |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| CO1            | Enumerate the different classes of computers and where they are used in everyday life.                                                      | K2                           |
| CO2            | Compute the effect of hardware/software enhancements on the speedup of a processor using Amdahl's law.                                      | K3                           |
| CO3            | Interpret possible dependencies that can cause hazards in a given block of code.                                                            | K3                           |
| CO4            | Summarize different strategies followed to ensure Instruction Level Parallelism.                                                            | K2                           |
| CO5            | Compare different strategies followed to ensure Instruction Level Parallelism and different strategies followed to ensure Data Parallelism. | K3                           |
| CO6            | Illustrate the need for memory consistency models and cache coherence protocols and explain the principle behind it.                        | K3                           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 3   |     |     |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |
| <b>CO5</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |
| <b>CO6</b> | 3   | 3   | 3   | 3   |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                                              |                               |                              |                         |
|-------------------|------------------------------------------------------------------------------|-------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                                     | <b>Name of the Author/s</b>   | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Computer architecture: A Quantitative Approach.                              | Hennessy, J. and Patterson, D | Morgan Kaufman               | 5/e, 2012               |
| 2                 | The Dark Side of Silicon: Energy Efficient Computing in the Dark Silicon Era | Kanduri, Anil, et al.         | Springer                     | 1/e, 2017               |

| <b>Reference Books</b> |                                 |                                    |                              |                         |
|------------------------|---------------------------------|------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>        | <b>Name of the Author/s</b>        | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Computer Architecture           | Gérard Blanchet<br>Bertrand Dupouy | Wiley                        | 1/e, 2013               |
| 2                      | Advanced Computer Architectures | Sajjan C Shiva                     | Taylor & Fancis              | 1/e, 2018               |
| 3                      | Computer Architecture           | Charles Fox                        | no starch press              | 1/e, 2024               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                              |
| <b>1</b>                              | <a href="https://archive.nptel.ac.in/courses/106/103/106103206/">https://archive.nptel.ac.in/courses/106/103/106103206/</a> |